BRS: Branching and Recursive Sampling for Motion Planning of Autonomous Mining Trucks in Narrow Unstructured Loading Pockets

Bai Li, Fan Bu, Guizhen Yu, and Zhijiang Shao

***Abstract*—Parking a large mining truck inside a narrow loading pocket requires accurate terminal alignment and economical maneuvers. This paper presents branching and recursive sampling (BRS), a deterministic motion planner for relatively open approaches and recessed destinations. Three complementary constructions generate reverse-only direct and layered paths, goal-side reverse entries, and start-side forward departures. The reverse-only branch returns its first feasible trajectory. The one-cusp branches retain the lowest complete execution cost from bounded candidate selection. When local constructions fail, reachable prefixes create new starts for the same solver. A common time parameterization enforces acceleration, lateral-acceleration, and steering-rate limits and accounts for gear changes and stationary steering. Evaluation on 100 fixed mining-map-derived and constructed tasks uses four native C++ search baselines, including a representation-controlled continuous-curvature A* planner. Under a common 10 s query limit, BRS solves all 100 tasks, with mean planning time 0.411 s and maximum 8.09 s. It attains the lowest comparison cost on 51 tasks including ties, with mean predicted execution time 67.84 s and 0.47 extra stops over the common-success set. Finite-selection, graph-search, and bounded-extension properties are established under explicit assumptions.**

***Index Terms*—Autonomous mining trucks, motion planning, deterministic sampling, recursive planning, continuous curvature.**

I. Introduction

Autonomous mining trucks (AMTs) are increasingly used in surface-mine transport, where productivity, vehicle utilization, and workforce exposure jointly shape deployment [1]. Early field-robotics research identified the integration of autonomous equipment with mine operations as a central challenge [2]. The Mining 5.0 perspective connects equipment autonomy with coordinated, human-centered mining systems [3]. Chen *et al.* demonstrate cooperative driving, loading, and unloading within an integrated autonomous mining system [4]. Precise forward/reverse tracking and fixed-point stopping have also been demonstrated on heavy-duty mining trucks by Hu's group [5]. Loading-site parking motivates dedicated maneuver planning [6]. Earlier work further formulates mining-truck loading and dumping through mixed-integer trajectory planning [7].

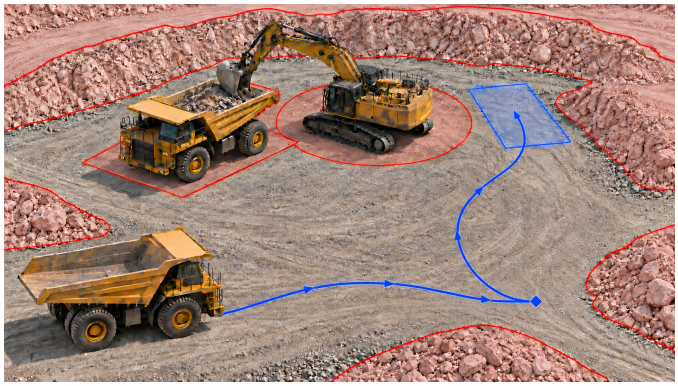


Fig. 1. Motion planning in a narrow loading pocket; red regions are non-traversable.

This paper focuses on narrow unstructured loading pockets, illustrated in Fig. 1. An arriving truck must reverse into a recessed loading position while avoiding irregular banks, scattered material, the excavator's rotational clearance, and a truck already being loaded. The approach is relatively open, but its available adjustment space remains limited. The long truck body and inward-extending pocket can require several maneuvers. Excavation changes the working boundary between operations, so a fixed offline trajectory cannot serve every task. We generate motion against the current static map and require an accurately aligned final reverse approach.

The practical objective extends beyond geometric length. Additional braking, restarts, and reversals prolong the loading cycle. A curvature jump within a single-direction path also requires a feasible steering transition, potentially at rest. Execution time and additional stops therefore matter even when a modest difference in planning latency is acceptable.

We propose branching and recursive sampling (BRS), a deterministic motion planner. Three complementary constructions cover reverse-only motion, goal-side reverse entry, and start-side forward departure. They share continuous-curvature interfaces. Reverse-only planning returns immediately; one-cusp constructions compare bounded shortlists through model-predicted execution time and extra stops. Bounded prefix expansion relocates the starting state and reuses the same constructions when local connections fail. The contribution is this coordinated maneuver representation and execution-aware selection, supported by finite-selection properties and

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comparisons using a shared curve representation.

II. Related Work

Table I organizes representative methods by curve construction, search, selection, and timing. Section VIII compares search implementations with both piecewise-circular and continuous-curvature representations.

A. Direct Optimization and Collision Representations

Optimization-based collision avoidance (OBCA) [8] reformulates collision constraints through convex duality. Finite unions of convex sets represent full-dimensional vehicles, obstacles, and irregular mine boundaries. Its parking example uses a Hybrid A* warm start and optimizes trajectories with motion and steering-rate constraints. A complex boundary may require many convex components and associated dual variables at multiple time samples. The resulting representation size affects solver workload; the collision reformulation alone does not establish planner completeness.

B. Search and Sampling With Vehicle Constraints

Rapidly-exploring random tree star (RRT*) [15] incrementally samples the free state space, chooses a feasible parent with minimum accumulated cost, and rewires nearby vertices when the new connection reduces their cost. Its probabilistic completeness and asymptotic optimality are proved under sampling, connection-radius, clearance, and cost assumptions. The foundational analysis concerns geometric paths without differential constraints; applying it to mining trucks requires a compatible steering connector, direction-aware states, and terminal-pose handling. BRS instead ranks deterministic maneuver families and expands starting configurations through recursive prefixes.

The Hybrid A* (HA) framework of Dolgov *et al.* [9] keeps continuous vehicle poses in discretized search cells, penalizes reverse travel and direction changes, and improves the search result through nonlinear smoothing. The full framework is therefore a search–optimization combination. The search accommodates obstacle layouts and several reversals, while state merging limits optimality guarantees. Raw arc transitions require steering treatment before execution.

Closed-loop rapidly-exploring random trees (CL-RRT) [16] sample reference commands and propagate a vehicle with its feedback controller. The resulting tree represents predicted closed-loop motion, with environment-dependent sampling, feasibility tests, and continuing improvement. This supports dynamically feasible maneuvers and safe stopping. Applying it to exact pocket parking requires a suitable reversing controller and terminal connection or acceptance rule.

State lattices (SL) [11] connect sampled states through feasible motion primitives, with relevant variables such as heading and curvature included in the state. Compatible adjoining primitives avoid discontinuities during reconnection. Resolution-related guarantees depend on the control-set construction and systematic search. Our first branch uses a task-aligned layered graph with zero-curvature interfaces, rather than a translationally repeated lattice over the entire configuration space. Its guarantees therefore depend on that restricted edge set.

Wang *et al.* [12] develop a bidirectional improved A-search guided tree (BIAGT). Two trees exchange arrival-cost information, while prioritized primitive modes allocate expansion effort. The full architecture also schedules passage through moving-obstacle conflicts; we consider its static planning component. Their completeness argument assumes an adequate primitive representation, eventual exploration of necessary modes, and arbitrarily sufficient resources. Distance-based rejection and heuristic inflation further affect finite-budget behavior.

C. Continuous-Curvature and Mining-Pocket Planning

Fraichard and Scheuer's continuous-curvature steering method (CC-Steer) [10] replaces instantaneous curvature changes with clothoid transitions and bounds sharpness. It supports forward and reverse travel and can serve as a steering component inside a general planner. Its topological-admissibility result concerns the steering method and compatible planning schemes. This establishes the curve vocabulary used here. BRS contributes a task-specific organization of candidate generation and continuation; the continuous-curvature A* comparison uses the same connector to examine that organization separately from curve smoothness.

Li *et al.*'s human-like parking planner (HLPP) [6] directly addresses individual and queued mine-loading tasks. It learns rewards for safety, travel time, terminal maneuver length, and courtesy from driving demonstrations. Its queued planner searches outward from the loading pose for a waiting configuration, then connects the arrival state using Hybrid A*. Failed waiting configurations trigger reselection; clustering supports exploration beyond repeatedly unsuccessful regions. An iterative optimization stage smooths curvature within corridors, followed by a learned steering–speed mapping.

The shared application makes HLPP a close reference. BRS defines separate reverse-only, goal-side, and start-side motion families with zero-curvature interfaces, compares model-predicted execution costs, and carries dynamically reachable prefixes into repeated local calls. Its terminal reverse segment is a hard geometric requirement. HLPP additionally models the preceding truck's departure and learned waiting behavior. The present static task fixes parked vehicles as obstacles and evaluates maneuver generation and continuation under one explicit vehicle model.

D. Combined Planning and Optimization

Bergman *et al.* [13] tightly couple a lattice planner and numerical optimal control by matching their vehicle models and objective functions; this combination is denoted Lattice–OCP. The lattice resolves combinatorial choices and supplies an informed initialization; continuous optimization then improves the trajectory beyond the discrete representation. Their analysis distinguishes sampled-family optimality from optimality in the original continuous problem.

TABLE I
CURVE CONSTRUCTION AND PLANNING MECHANISMS

Method	Curve model	Exploration	Selection	Timing
HA [9] (2010)	Arcs, smoothing	A* cells	Weighted length	Path output
CC-Steer [10] (2004)	Clothoids, arcs	Steering connector	Geometric construction	Sharpness bound
OBCA [8] (2021)	Vehicle dynamics	Nonlinear program	Trajectory cost	Joint optimization
SL [11] (2009)	Motion primitives	State lattice	Graph cost	Primitive dependent
BIAGT [12] (2024)	Circular primitives	Two trees	Tree cost	Conflict scheduling
Lattice–OCP [13] (2021)	Primitives, OCP	Lattice, optimizer	Shared objective	Joint optimization
DFTPAV [14] (2024)	Flat polynomials	HA, optimizer	Trajectory cost	Joint optimization
HLPP [6] (2025)	RS, curvature smoothing	Waiting-point search	Learned reward	Learned speed map
C-MINLP [7] (2023)	Arcs, OCP	GHA, optimizer	Trajectory cost	Speed assignment
BRS	Clothoids, arcs	Three branches, prefixes	Common execution cost	Rate-limited profile

Han *et al.* [14] develop a differential-flatness trajectory planner for autonomous vehicles (DFTPAV), combining a Hybrid A* front end with spatial–temporal trajectory optimization using differential flatness and piecewise polynomials. Safe driving corridors support static collision avoidance, while moving objects receive a separate distance treatment. Gear-segment directions are fixed from the front end, although junction positions and orientations can be optimized. Their framework jointly optimizes timing and geometry. Transfer to the present benchmark would require identical full-stop boundary conditions and steering constraints, including careful treatment of the flatness singularity near zero speed.

Our earlier conditional mixed-integer nonlinear programming (C-MINLP) framework [7] addresses mining-truck loading and dumping through generalized Hybrid A* (GHA), speed assignment, and conditional nonlinear optimization. It explores maneuver alternatives and then optimizes continuous variables. BRS generates its final geometry directly through continuous-curvature families and layered connections. Both approaches benefit from scenario structure; their candidate organization, refinement operation, and execution-cost selection differ. The experiments include GHA-search under the common evaluator and a separate continuous-curvature A* search.

III. Problem Formulation

A. Static Task and Vehicle Model

A task comprises a static planar map, a start pose $q_s = (x_s, y_s, \theta_s)$, and a goal pose $q_g = (x_g, y_g, \theta_g)$. Pose refers to the rear-axle center. The prescribed terminal approach is a reverse straight segment of length at least ℓ_g aligned with θ_g. Define its upstream anchor

$$A = (x_g + \ell_g \cos\theta_g,\ y_g + \ell_g \sin\theta_g,\ \theta_g). \tag{1}$$

The map is held fixed during each planning query. The task models low-speed maneuvering in a plane.

Let $\nu \geq 0$ denote speed magnitude, $g \in \{-1, +1\}$ the reverse/forward gear, δ the front-wheel steering angle, and L_w the wheelbase. The state $z = (x, y, \theta, \nu, \delta)$ follows

$$\begin{aligned} &\dot{x} = g\nu\cos\theta, \quad \dot{y} = g\nu\sin\theta, \\ &\dot{\theta} = g\nu\tan\delta/L_w, \quad \dot{\nu} = a, \quad \dot{\delta} = \omega. \end{aligned} \tag{2}$$

The controls satisfy

$$\begin{aligned} &0 \leq \nu \leq \bar{v}_g, \quad -\bar{a}_b \leq a \leq \bar{a}_f, \\ &|\delta| \leq \bar{\delta}, \quad |\omega| \leq \bar{\omega}, \quad \nu^2|\kappa| \leq \bar{a}_y, \qquad \kappa = \tan\delta/L_w. \end{aligned} \tag{3}$$

Here $\bar{a}_f$ and $\bar{a}_b$ bound acceleration and braking of speed magnitude. A gear change is permitted only at rest. The start and goal conditions are

$$z(0) = (q_s, 0, 0), \qquad z(T_f) = (q_g, 0, 0). \tag{4}$$

Numerical pose tolerances, the reverse terminal segment, and the initial/final steering convention are identical for every tested method.

B. Body, Planning Contour, and Feasibility

Let l_f, l_r be the distances from the rear axle to the front and rear body extremities and w the physical width. With directional buffers $b_f, b_r, b_l, b_{\text{rt}}$, the planning rectangle in the body frame is

$$\mathcal{B}_b = [-l_r - b_r, l_f + b_f] \times [-w/2 - b_{\text{rt}}, w/2 + b_l]. \tag{5}$$

Its world footprint is $\mathcal{F}_b(q) = p + R(\theta)\mathcal{B}_b$. The physical body is obtained by setting the buffers to zero. Solid and dashed vehicle outlines in the trajectory plots denote the physical body and buffered planning contour, respectively. For allowed workspace $\mathcal{H}$ and occupied map set $\mathcal{O}$, the collision constraints are

$$\mathcal{F}_b(q(t)) \subseteq \mathcal{H}, \qquad \mathcal{F}_b(q(t)) \cap \mathcal{O} = \varnothing, \quad t \in [0, T_f]. \tag{6}$$

The benchmark fixes the representation of $\mathcal{H}$ and $\mathcal{O}$. Circular exclusion regions represent the excavator's rotational clearance. Section VIII-A specifies the static boundary samples, obstacle representation, buffers, and validation resolution used in the experiments.

C. Execution Objective and Internal Stops

The common vehicle model predicts trajectory execution time from moving intervals and stationary steering or gear-change intervals. Let N_c count sign changes of g, and let N_s count extra internal stop events. An event simultaneously performing a gear change and a steering adjustment counts once. The common initial and terminal stationary states do

not contribute to N_s. A same-direction curvature jump may contribute a stop even though it does not increase N_c.

The target performance measure is

$$\min_{T_f, z(\cdot), a(\cdot), \omega(\cdot), g(\cdot)} J = T_f + \lambda_s N_s, \tag{7}$$

subject to (2)–(6), the terminal reverse approach, and stationary transition rules. λ_s is an equivalent-time operating penalty per extra stop. Real steering, shifting, and settling durations are already included in T_f; $\lambda_s N_s$ expresses the operating preference for fewer internal stops.

BRS follows a prioritized sequence of geometric constructions, validates complete trajectories, and applies the common time parameterization. Branch I returns its first feasible trajectory. Within a successful one-cusp construction, the selector retains the smallest J among the candidates actually evaluated within its finite budget. Geometric length remains the additive objective of the layered graph and the ordering criterion for one-cusp candidate screening.

IV. Overall Architecture and Common Operations

A. Branch Priority and Acceptance

Figure 2 summarizes the solver. Branch I attempts a reverse-only connection with $N_c = 0$. Branch II grows a reverse-entry family outward from the goal and connects the start to a selected cusp. Branch III grows a forward family from the start and connects the cusp back to the goal approach. Both latter branches contain at most one forward-to-reverse cusp before prefix concatenation.

A direct catalogue supplies reverse-only connections and the zero-fan limit of the goal-side family. It joins the start to the terminal anchor using the shared continuous-curvature connector, followed by the compulsory reverse straight. Branch I checks its reverse catalogue in length order and returns the first complete feasible trajectory. Layered search is invoked when that catalogue fails.

The wrapped heading difference $\Delta = |\operatorname{wrap}(\theta_s - \theta_g)|$ determines the order of the two one-cusp constructions:

$$\pi(\Delta) = \begin{cases} (\mathrm{II}, \mathrm{III}), & \Delta \le \alpha, \\ (\mathrm{III}, \mathrm{II}), & \Delta > \alpha. \end{cases} \tag{8}$$

When Branch I fails, the solver follows $\pi(\Delta)$ at the current sampling level. Branch II first checks its direct forward-entry catalogue and then its reverse-entry fans; Branch III checks its forward-departure fans. The next branch is entered only if the current construction produces no complete feasible trajectory. If both one-cusp constructions fail, their next sampling level is explored.

For a successful Branch II or III construction, the first fully checked, time-parameterized trajectory becomes the incumbent. The same construction may evaluate additional candidates until either K_f feasible candidates have been examined or a post-solution budget τ_{sel} expires. A replacement requires a strictly smaller complete-trajectory J beyond tolerance τ_J. The solver then returns this incumbent without starting another branch. Branch I returns immediately. Optional local refinement has its own bounded budget and the same incumbent-retention rule.

Every accepted trajectory satisfies body-footprint, curvature, endpoint, junction, and terminal-approach constraints. A shared monotonic-clock deadline $\tau_{\max}$ covers map indexing, curve generation, graph search, collision checking, child invocations, and time parameterization. Deadline checks occur before new operations and within long loops. A return reserve τ_{ret} covers final assembly and resource release. Completed incumbents survive interruption of subsequent candidate evaluation; a query without one reports failure.

B. Continuous-Curvature Turn and Connector

The shared curve representation consists of straight lines, circular portions, and linear-curvature ramps. Continuous-curvature constructions extend the bounded-curvature path vocabulary [10]. Every generated turn begins and ends with zero curvature, so neighboring turns and straight segments join without a steering-angle jump.

For signed heading change ϕ, nominal radius R, and ramp parameter ℓ_r, set

$$\begin{aligned} \sigma(\phi) &= \frac{1}{R\ell_r} \min\left(1, \sqrt{\frac{|\phi|}{\phi_\epsilon}}\right), \qquad k_p = \min\left(\frac{1}{R}, \sqrt{|\phi|\sigma}\right), \\ L_r &= \frac{k_p}{\sigma}, \qquad L_a = \max\left(0, \frac{|\phi| - k_p L_r}{k_p}\right). \end{aligned} \tag{9}$$

The parameter ϕ_ϵ defines a small-angle regularization range. At $\phi = 0$, the turn has zero length and curvature. For nonzero angles approaching zero, both the peak curvature and sharpness vanish, so an arbitrarily small turn does not impose a finite steering-rate speed cap. For $\phi \neq 0$, the curvature ramps from zero to $\operatorname{sgn}(\phi) k_p$ over L_r, remains constant for L_a, and returns to zero over L_r. A short turn has a triangular curvature profile and need not reach $1/R$; thus R is a curvature-scale parameter, not necessarily the radius attained by every candidate.

For poses q_a, q_b, a connector contains a first turn, a tangent straight of length d, and a second turn. Let β be the world heading of that straight. With turning signs $s_1, s_2 \in \{-1, +1\}$, the heading changes are

$$\begin{aligned} \phi_1(\beta) &= s_1 \bmod_{2\pi}\{s_1(\beta - \theta_a)\}, \\ \phi_2(\beta) &= s_2 \bmod_{2\pi}\{s_2(\theta_b - \beta)\}. \end{aligned} \tag{10}$$

Let $U(\phi; R, \ell_r)$ denote the complex displacement of a turn starting at heading zero and let $\Delta p = (x_b - x_a) + \mathrm{i}(y_b - y_a)$. Define

$$\begin{aligned} W(\beta) &= \left[\Delta p - e^{\mathrm{i}\theta_a} U(\phi_1) - e^{\mathrm{i}\beta} U(\phi_2)\right] e^{-\mathrm{i}\beta}, \\ \operatorname{Im} W(\beta) &= 0, \qquad d = \operatorname{Re} W(\beta) \ge 0. \end{aligned} \tag{11}$$

Equation (11) is a scalar root problem. A finite heading scan brackets roots separately between angle-wrap breakpoints; endpoint headings and the chord heading are also checked to retain degenerate zero-turn cases. Exact endpoint-heading roots are retained before numerical roots. Proximity-based deduplication keeps the original root values. Candidates exceeding the turn limit are discarded, and the integrated endpoint is verified. The finite scan defines the candidate

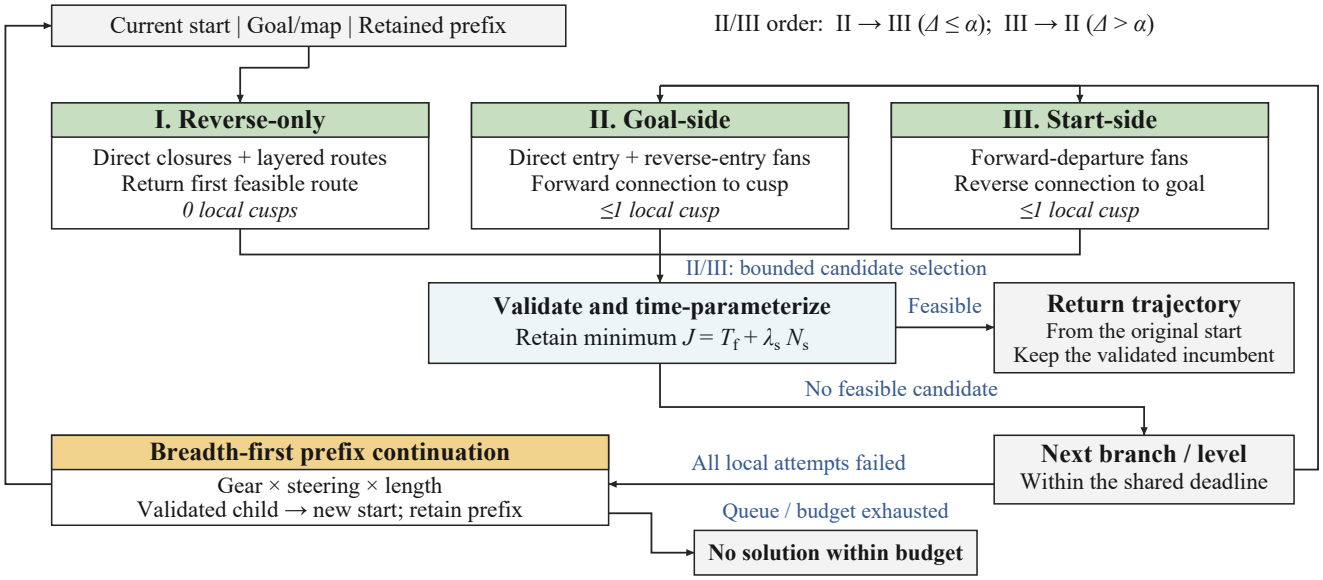


Fig. 2. Three prioritized constructions, bounded candidate selection, and prefix continuation under a shared deadline.

catalogue. Scalar root finding and quadrature construct each member.

Figure 3 shows an actual connector, with β, R, d, and the ramp identified directly. Interior graph edges use the same closure with signed heading differences confined to the local graph frame. Their candidate lists are sorted by complete curve length.

C. Swept-Footprint Checking

Curves are sampled in arc length with maximum spacing h_c. To cover motion between samples, let ρ bound the distance from the rear axle to any point of the planning contour and let K bound absolute curvature on the interval. An additional numerical checking allowance is

$$\varepsilon_h = \frac{h_c}{2}(1 + \rho K) + \varepsilon_{\text{num}}. \tag{12}$$

The footprint at each sample is enlarged by ε_h for checking. This allowance covers between-sample translation and rotation and is separate from the operational buffer. A coarse subset rejects collisions early; acceptance uses the full checking mesh. A spatial bounding-box hierarchy accelerates exact occupied-sample queries. Exact-input memoization reuses completed curve checks within a query. Each final trajectory also undergoes a finer-mesh validation.

Map discretization has a separate role. Let P be occupied samples and suppose a continuous occupied set obeys $\mathcal{O} \subseteq P \oplus B_{\varepsilon_m}$, where B_{ε_m} is a disk of radius ε_m. A sampled footprint enlarged by $\varepsilon_h + \varepsilon_m$ then certifies avoidance of $\mathcal{O}$ over its adjacent path interval. For a sampled straight boundary segment, half the maximum sample spacing bounds its covering radius; curved segments additionally require a chord-deviation bound. The benchmark supplies its fixed occupied samples and workspace. Boundary sampling gives a covering radius for the boundary itself.

D. Prefix Expansion and Return to the Entrance

When all local attempts fail, the solver explores a breadth-first queue of states carrying the current pose, complete prefix, depth, parent identity, last gear, and accumulated cusp count. From each parent, it applies each combination of forward/reverse and left/right steering at each length in a finite set $\mathcal{L}_p$. Each prefix ramps curvature from zero, follows a maximum-curvature turn when sufficient length remains, and ramps back to zero. For a prefix length $L \geq 2\ell_p$, its signed heading change is

$$\phi_p = s\, g\, (L - \ell_p)/R_{\min}. \tag{13}$$

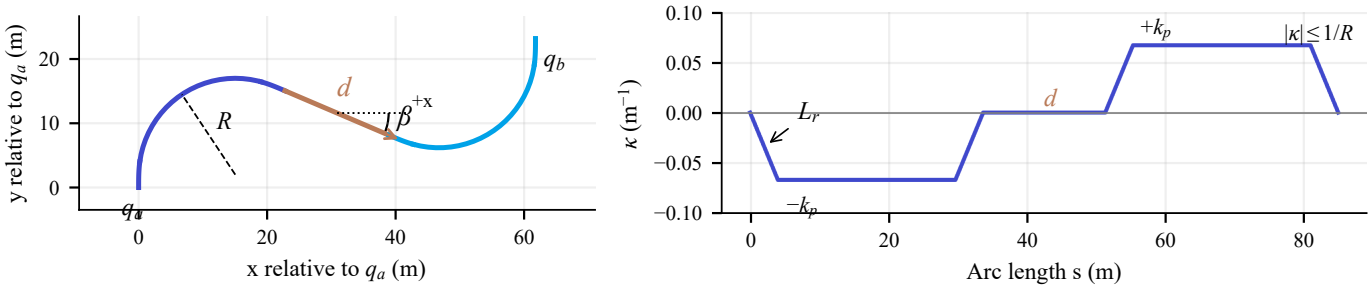
Prefixes that collide, exceed the cusp budget, or duplicate a stored state key are rejected. The key includes quantized pose, last gear, prefix cusp count, and depth. A surviving endpoint becomes the start of a new local invocation with the original map, goal, and vehicle. That invocation recomputes (8), begins again with Branch I, and uses its designated child sampling budget.

Each child candidate is evaluated together with its retained prefix from the original start. This complete-path check determines feasibility, the actual internal stops, and J. Once a child succeeds, bounded selection remains inside that child's successful construction. The concatenated incumbent is returned; other parent trials are not started. If the child fails, eligible states remain in the queue for further expansion. Termination occurs on success, an empty queue, the depth limit, the tested-state budget, or the shared deadline.

V. Branch I: Reverse-Only Planning

A. Reverse Construction and Task-Aligned Layers

The first branch constructs paths that can be executed entirely in reverse. It returns a feasible direct analytic closure immediately and invokes the layered graph when direct closure fails. Direct closures use the catalogue in Section IV; the graph



(a) Connector geometry (b) Curvature profile

Fig. 3. Continuous-curvature connector geometry and curvature profile, with the middle tangent and turning parameters identified.

construction below supplies paths with intermediate steering choices. Construction proceeds outward from q_g to A and then toward q_s; reversing this geometric traversal produces the actual motion $q_s \to A \to q_g$. The mandatory straight $q_g A$ is checked first. Its failure excludes every candidate in this branch under the prescribed terminal approach.

Let $D = \|p_s - p_A\|$ and let γ be the heading of the chord from A to q_s. In a frame with origin A and longitudinal axis γ, the graph contains the exact endpoint poses and $n-1$ interior layers, where $n = \max(1, \lceil D/h_x \rceil)$. Interior states combine longitudinal coordinate $x_j = jD/n$, lateral offsets $y \in \mathcal{Y}_c$, and headings $\psi \in \Psi_c$ relative to the chord. All graph interfaces have zero curvature. States whose vehicle footprints collide are discarded. The layered construction is enabled when $D > D_{\min}$ and $\max(|\psi_A|, |\psi_s|) < \psi_{\max}$ in this construction frame. Otherwise, a direct connector family is still attempted.

Edges join successive coarse layers. Edges satisfy $|y_v - y_u| \le \max(b_y, c_y(x_v - x_u))$ and $|\psi_v - \psi_u| \le \psi_e$ before curve generation. Layer order defines a directed acyclic graph (DAG). A direct endpoint-to-endpoint edge supplements the layered routes and uses several radius and ramp scales. Each ordinary edge uses the connector in Section IV-B with its designated scale. Multiple turns along successive edges allow a single reverse maneuver to pass obstacles on different sides.

B. Lazy Dynamic Programming

For each edge $e = (u, v)$, the connector produces a finite list $\mathcal{C}_e$ sorted by length. Only collision-free members can be used. Initially, an ungenerated edge has the admissible lower bound $w_e = \|p_v - p_u\|$. Bellman's recurrence in topological order is

$$V(v) = \min_{e=(u,v)} \{V(u) + w_e\}, \qquad V(A) = 0. \tag{14}$$

The lowest-cost tentative route is traced backward. Unknown edges on that route are generated and assigned their shortest candidate lengths. On the next relevant iteration, candidates awaiting validation are footprint-checked. A rejected candidate advances its edge to the next entry in $\mathcal{C}_e$; an exhausted edge receives infinite weight. Equation (14) is recomputed after these updates. A route is accepted only when every selected edge is validated. Generation and validation both consume the evaluation budget.

Equation (14) accumulates length, giving a shortest geometric route in its generated graph under the Appendix conditions. Time parameterization couples adjacent segments through boundary speeds, so complete-route execution cost is evaluated after graph search.

C. Refinement With Incumbent Retention

An optional refinement call retains a completed coarse trajectory, doubles the layer density, and samples lateral and heading offsets around the interpolated coarse route. Original coarse states are reinserted, and fine edges can span one or two fine layers. The corridor concentrates sampling near the coarse route. Fine search uses the same lazy recurrence with a post-solution budget τ_{fine} contained within the shared query deadline. The default first-feasible mode returns the coarse trajectory immediately; the refinement component is evaluated separately. The coarse solution remains an explicit incumbent. Both complete routes include the mandatory terminal straight and receive the common time parameterization. The fine route replaces the coarse route only when its J is strictly smaller beyond numerical tolerance. The selected full path is rechecked. Consequently, refinement cannot increase the selected execution cost, even when a shorter geometric route requires slower steering transitions.

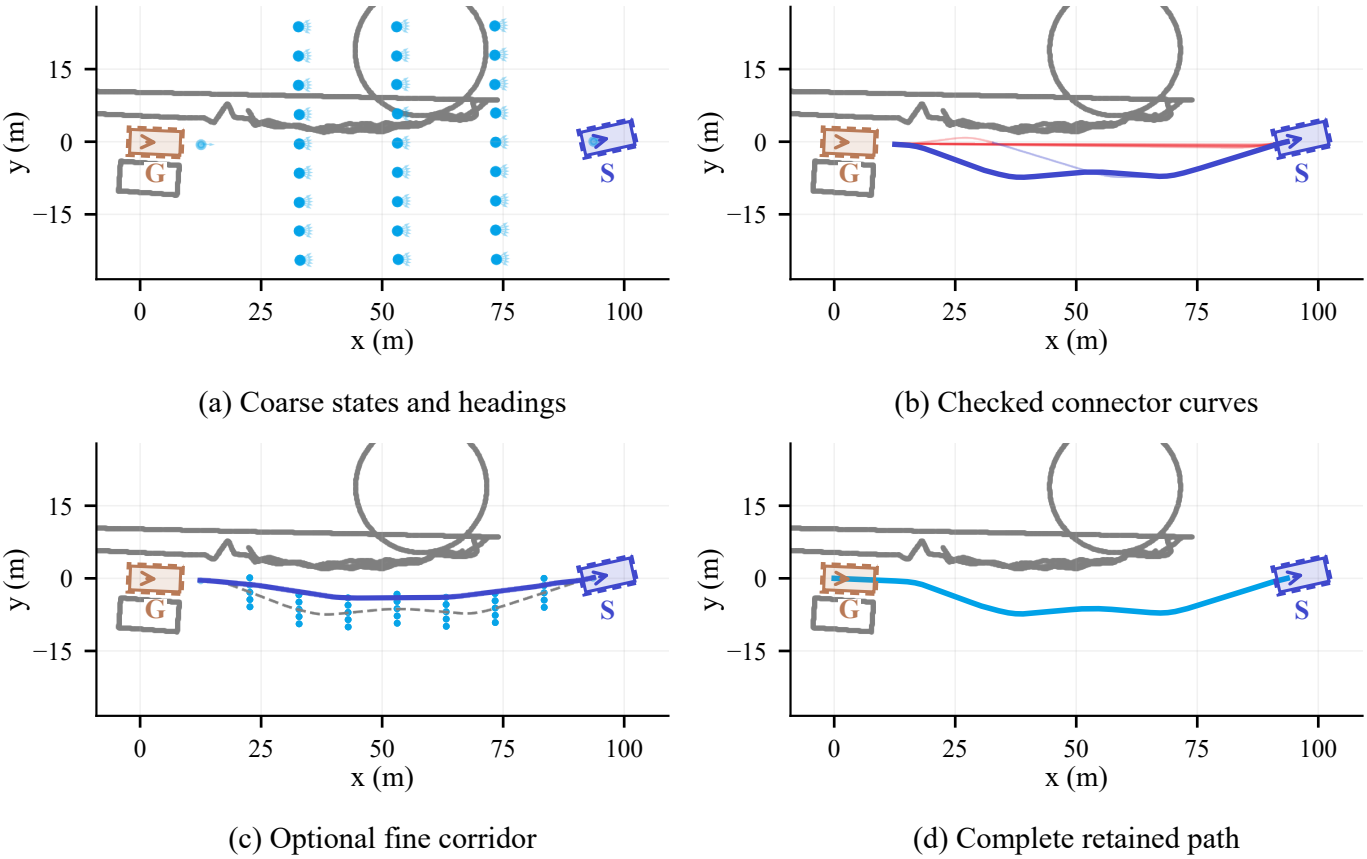
Figure 4 separates state placement, generated curves, optional local refinement, and the retained layered path. Zero curvature at every graph interface prevents a steering jump between edges. Curvature derivatives may change at primitive boundaries; the common timing procedure accounts for their finite steering-rate requirements.

VI. Branch II: Goal-Side Reverse-Entry Families

Branch II asks where the truck can begin a collision-free reverse entry into the pocket. It exploits the goal's restricted geometry first, before connecting to the more open start region. For each terminal-stem extension $o \in \mathcal{O}_g$, define

$$A_o = p_g + (\ell_g + o)e(\theta_g), \qquad e(\theta) = (\cos\theta, \sin\theta). \tag{15}$$

The angle set includes zero, retaining a straight-entry limit. From (A_o, θ_g), outward construction applies each turn sign $s \in \{-1, +1\}$, angle $\phi \in \Phi_k$, radius $R \in \mathcal{R}_k$, and ramp $\ell_r \in \mathcal{H}_k$. A straight stem followed by this continuous-curvature turn ends at a candidate cusp pose q_c. Its construction direction is



(a) Coarse states and headings

(b) Checked connector curves

(c) Optional fine corridor

(d) Complete retained path

Fig. 4. Layered reverse construction: coarse states, tested curves, refined sampling, and the retained route.

opposite to its later execution: reversing the fan yields the feasible-entry portion $q_c \to q_g$.

The complete fan is collision-checked before attempting the remaining connection. For each surviving fan, the connector generates forward paths $q_s \to q_c$. The endpoint body heading agrees with that of the reverse fan, and both sides have zero curvature at q_c. Thus the cusp changes travel direction while retaining a continuous body pose and a compatible steering state. The complete maneuver comprises one forward leg and one reverse leg, including the required final straight.

Let L_F and L_R be its forward and reverse lengths. Complete candidates are ordered by

$$C_{\text{geo}} = L_F + w_R L_R. \tag{16}$$

A priority queue realizes this order without generating every connector in advance. For a safe goal-side fan of length L_R ending at q_c, its unresolved entry has lower bound $\|p_s - p_c\| + w_R L_R$. Popping that entry generates its finite connector catalogue and inserts complete candidate costs. Popping a complete entry triggers connector and full-trajectory validation. Generation order breaks equal-cost ties. Thus an unresolved fan cannot conceal a lower-cost complete candidate when a complete entry is selected.

The first validated candidate supplies an executable incumbent. At most K_f feasible candidates are admitted within the additional budget τ_{sel}, and their complete speed profiles determine J. The best evaluated incumbent is retained. For a recursive call, scoring includes the retained prefix. If no complete candidate succeeds, control follows the branch and level transitions in Fig. 2.

The levels broaden the representation in two ways. An intermediate level adds larger turning radii. A later level uses denser turn angles and shorter curvature ramps, together with its designated connector scales. Larger radii provide gentler lateral excursions; denser angles expose narrower admissible cusp regions. The top row of Fig. 5 shows the complete generated fan set, the collision-free subset, and the selected maneuver for a goal-side example.

VII. Branch III and Common Trajectory Generation

A. Start-Side Forward Families

Branch III changes which side determines the cusp. It first uses the available start-side turning room: for $o \in \mathcal{O}_s$, a forward straight from q_s of length o is followed by a turn sampled from the same sign, radius, ramp, and angle sets. This produces a forward fan ending at q_c. Colliding fans are rejected immediately.

The goal-side anchor remains A from (1). A connector is constructed outward from A to q_c and then traversed in reverse, yielding $q_c \to A \to q_g$. This is distinct from Branch II, where the admissible reverse-entry fan is established first and the connector lies on the forward leg. The two constructions can therefore reach different cusp regions even when they use identical scale sets.

An unresolved start-side fan of length L_F uses the lower bound $L_F + w_R(\|p_A - p_c\| + \ell_g)$. The same priority queue generates and checks complete candidates in (16) order. Bounded selection retains the smallest evaluated complete-path J. Both families use at most one cusp locally; a zero-length forward leg produces a reverse-only limiting case. A retained prefix can add cusps or join in the same direction, so the final count is recomputed after concatenation. The bottom row of Fig. 5 illustrates the start-side construction.

B. Path-to-Trajectory Conversion

All geometric planners use one path–velocity decomposition [17]. A path is split into moving blocks at gear changes and retained curvature jumps. Let s denote traveled distance and let $\kappa(s)$ be physical steering curvature, with the gear transformation already applied. Steering is $\delta(s) = \arctan(L_w \kappa(s))$. On a continuous block the speed ceiling is

$$v_{\max}(s) = \min\left\{\bar{v}_g, \sqrt{\frac{\bar{a}_y}{|\kappa(s)|}}, \frac{\bar{\omega}}{|\delta'(s)|}\right\}. \tag{17}$$

The steering derivative is

$$\delta'(s) = \frac{L_w \kappa'(s)}{1 + (L_w \kappa(s))^2}. \tag{18}$$

A zero denominator gives an inactive bound. The implementation bounds curvature and steering derivative over each complete interval. For linear-curvature pieces these extrema follow from the endpoint curvature values and the minimum absolute curvature in the interval.

The block is sampled in arc length. Starting and ending speeds are zero at required stops. A forward pass enforces $v_{j+1}^2 \le v_j^2 + 2\bar{a}_f \Delta s_j$; a backward pass enforces $v_j^2 \le v_{j+1}^2 + 2\bar{a}_b \Delta s_j$. Both respect the interval speed ceilings. Under constant acceleration within each interval,

$$\Delta t_j = \frac{2\Delta s_j}{v_j + v_{j+1}}. \tag{19}$$

This forward–backward pass supplies the common model-predicted execution time.

At an internal event, let $\Delta\delta$ be the change between the incident steering angles and I_g indicate a gear change. The stationary duration is

$$\tau_{\text{event}} = \max\left(\frac{|\Delta\delta|}{\bar{\omega}},\ I_g \tau_g\right) + \tau_0. \tag{20}$$

The execution model allows concurrent gear selection and stationary steering, followed by settling time τ_0. This common simulation convention defines the predicted transition duration. A same-gear curvature jump still creates a stop and a steering duration. A combined gear and steering change counts as one event. Any steering adjustment required at the initially and finally stationary poses contributes to T_f but not to the count of extra internal stops N_s.

Continuous-curvature paths receive steering-rate speed limits throughout each moving leg. Piecewise-circular paths additionally receive stationary steering events at retained curvature jumps. The same evaluator applies to BRS and every geometric baseline. This supports a comparison of complete model-feasible trajectories, including speed reduction, stopping, steering, and gear changes.

TABLE II
COMMON PHYSICAL, TIMING, AND VALIDATION PARAMETERS

Parameter	Value
Body length $l_f + l_r$; width w	10.4; 4.9 m
Front/rear axle offsets l_f, l_r	8.32; 2.08 m
Wheelbase L_w; minimum radius $R_{\min}$	6.22; 15 m
Buffers $b_f, b_r, b_l, b_{\text{rt}}$	0.5 m each
Final reverse straight ℓ_g	12.54 m
Forward/reverse limits $\bar{v}_+, \bar{v}_-$	4; 2 m/s
Acceleration/braking $\bar{a}_f, \bar{a}_b$	0.5; 0.7 m/s^2
Lateral acceleration $\bar{a}_y$	1 m/s^2
Steering rate $\bar{\omega}$	5 deg/s
Shift τ_g; settling τ_0	2; 0.5 s
Extra-stop evaluation weight λ_s	10 s/stop
Timing / planning check spacing	0.1 / 0.04 m
Independent check spacing	0.02 m
Position / heading tolerance	10^{-5} m / 10^{-7} rad
Join curvature tolerance	10^{-8} m^{-1}
Steering-jump detection tolerance	10^{-9} m^{-1}
Obstacle-body filling pitch	0.25 m
Planning limit / return reserve	10 / 0.2 s

VIII. EXPERIMENTS

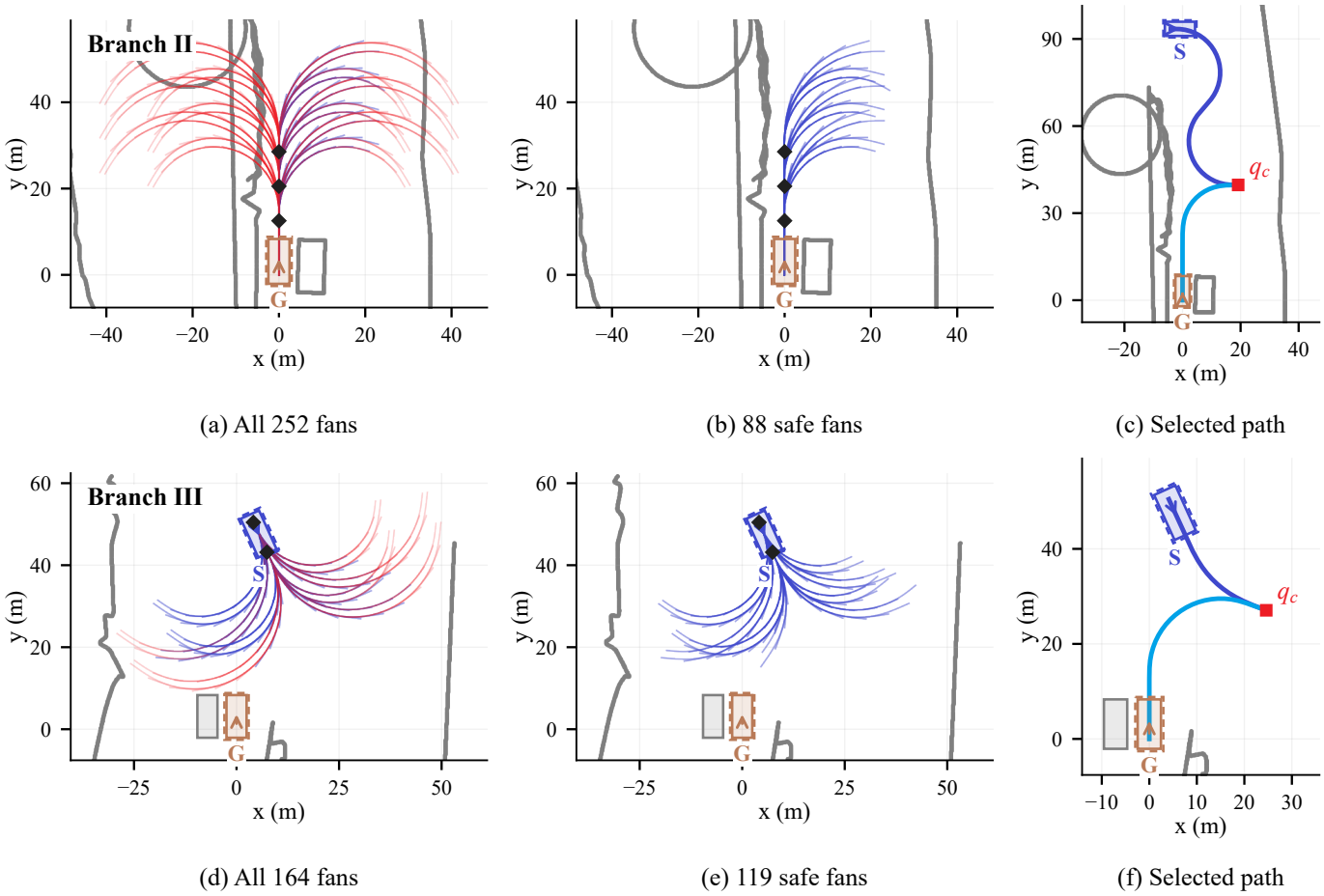
A. Benchmark and Common Settings

The main benchmark contains 100 fixed planar tasks, Cases #1–#100. Ninety-one use mining-map-derived geometry with controlled start/goal tasks, eight use constructed environments, and one contains a documented corrected starting pose. Six source groups contribute 48, 28, 11, 9, 3, and 1 tasks. Coordinates are local and metric; plots use rigidly transformed frames to enlarge the maneuvering region. Task provenance and input hashes are released with the dataset.

The static map comprises occupied samples, filled parked-body interiors, and a convex-hull workspace. Table II separates physical dimensions and planning buffers. The footprint certificate applies to this representation under Section IV.

All five planners run as native C++ programs on Windows 11, using an Intel Core i9-12900H (14 cores, 20 logical processors) and 16 GB nominal RAM (15.75 GiB visible). The common Release build uses LLVM-MinGW Clang 23.1.4 with C++17, -O3 -DNDEBUG, and static linking. Each planner uses one computational thread pinned to logical processor 2, a performance core. The same executable reads each task and invokes the requested planner. Compiler identification, build flags, input hashes, and executable hashes accompany the results. Methods run sequentially, with their order rotated across tasks. A lightweight monitor records process activity during measurement.

Planning time is wall-clock time from query entry through complete validation, time parameterization, and return preparation. Scene-file loading, plotting, output serialization, and the additional finer-mesh audit are excluded. Every method has the same 10 s limit, enforced through the shared deadline. Results returned beyond the limit are failures. Each task–method pair contributes one run to the main comparison. Execution time is predicted by the shared vehicle model. Steering rate, shift duration, and settling time are common fixed simulation settings; simultaneous gear and steering adjustment uses their maximum duration plus settling.



(a) All 252 fans (b) 88 safe fans (c) Selected path

(d) All 164 fans (e) 119 safe fans (f) Selected path

Fig. 5. Goal-side and start-side families: generated fans, feasible fans, and selected motions.

B. Comparison Planners

Four search implementations share the vehicle, static maps, terminal reverse straight, collision rules, and execution evaluator.

Hybrid A (HA)* implements the search component of [9]. Forward and reverse circular primitives have length 3 m and curvature multipliers $\{-1, -0.5, 0, 0.5, 1\}/R_{\min}$. The stored key uses 1 m position cells, 5° heading bins, and gear. The heuristic combines obstacle-grid and Reeds–Shepp (RS) distances [18]. RS closure is attempted at the root and every ten expansions. HA returns its first validated result.

BIAGT implements the static tree-search mechanisms of [12]: shared tree costs, prioritized forward/reverse modes, distance-based rejection, and RS bridges between the trees. The metric is $d_R(q,p) = \sqrt{\|p_q - p_p\|^2 + R_{\min}^2 \,\mathrm{wrap}(\theta_q - \theta_p)^2}$. Rejection, cross-tree proximity, and cost-sharing radii are 1, 20, and 30 in this metric; heuristic inflation is 1.5. Shared costs examine opposite-tree states using lower-bound pruning. This experiment evaluates the static search under the common execution model.

GHA-search implements the search front end of C-MINLP [7]. It exchanges endpoint roles, uses a zero heuristic, and generates forward/reverse Dubins connections [19]. It gathers up to 100 candidates under its time budget and a 150,000-node limit. The smallest common J is retained. The geometric search uses reverse weight 1.35, switch penalty 5, and steering-change penalty 2.

Continuous-curvature A (CC-A*)* controls the curve representation in the comparison. It uses the same continuous-curvature turn, endpoint connector, and direct-closure scales as BRS, with conventional pose-bin A* exploration. Its key includes position, heading, gear, and cusp count. Zero-curvature-interface primitives have lengths $\{4, 8, 12\}$ m, directions $\{-1, +1\}$, and steering scales $\{-1, -0.5, 0, 0.5, 1\}$. The heuristic combines Euclidean, heading-change, and obstacle-grid estimates. At the root and every ten expansions, the connector tries both travel directions. RS words additionally propose cusp poses, with each same-direction section reconstructed by the common continuous-curvature connector. The first successful closure batch supplies the candidate shortlist; its smallest J is returned. The cusp cap is three. This baseline combines conventional search with the shared curve family, permitting a comparison of candidate organization beyond steering continuity.

All geometrically valid candidates use the same speed and stop evaluator. HA and BIAGT preserve their first-feasible stopping rules. GHA-search and CC-A* retain the smallest J among their admitted feasible candidates. BRS returns immediately in Branch I and uses bounded current-

TABLE III
PROPOSED PLANNER PARAMETERS; DISTANCES IN METERS AND ANGLES IN DEGREES UNLESS STATED OTHERWISE

Component	Values
Branch priority	$\alpha = 135^\circ$; root levels $K = 3$; child levels $K_{\text{child}} = 1$.
Coarse DAG	$h_x = 22$; $\mathcal{Y}_c = \{-24, -18, \dots, 24\}$; $\Psi_c = \{-30, -15, 0, 15, 30\}$. $D_{\min} = 6$; $\psi_{\max} = 85^\circ$; edge bounds $b_y = 6$, $c_y = 0.65$, and $\psi_e = 65^\circ$.
Direct closures	Radius factors $\{1, 1.3, 1.7\}$, ramps $\{4, 8\}$, turn limit 1.8π rad; first feasible reverse curve; bounded forward shortlist.
Graph connectors	Direct radii $\{1, 1.3, 1.7\}R_{\min}$, ramps $\{4, 8\}$, individual turn limit 0.98π rad. Interior edges: $R = R_{\min}$, ramp 4; closure scan at 41 evenly spaced headings in $[-\pi/2 + 0.02, \pi/2 - 0.02]$ rad plus endpoint/chord headings.
Fine DAG	Layer spacing halved; lateral offsets $\{-3, -1.5, 0, 1.5, 3\}$ and heading offsets $\{-15, -7.5, 0, 7.5, 15\}$ around coarse interpolation; one/two-layer edges; coarse/fine evaluation budgets 1800/700; optional refinement budget 0.2 s.
One-cusp level 1	Goal offsets $\mathcal{O}_g = \{0, 8, 16\}$; start offsets $\mathcal{O}_s = \{0, 8\}$. Fan radii $\{1, 1.4\}R_{\min}$, ramps $\{4, 8\}$, angles $\{0, 15, \dots, 150\}$. Connector radii match fan scales; connector ramp 4.
One-cusp levels 2, 3	Level 2 radii $\{1, 1.4, 1.8, 2.4\}R_{\min}$. Level 3 retains expanded fan radii, uses angles $\{0, 1, \dots, 150\}$ and ramp 1, and connector radii $\{1, 1.4\}R_{\min}$ with ramp 1.
General connector	Individual turn limit 1.8π rad; scan spacing at most $\pi/72$ rad between wrap breakpoints; scalar root tolerance 10^{-12}; root deduplication 10^{-10} rad. Interior-edge roots use 10^{-10} rad deduplication. Exact floating-point collision-cache keys.
Prefix expansion	$\mathcal{L}_p = \{4, 8, 12\}$, ramp $\ell_p = 1$, four gear/steering modes, depth limit 2, tested-state limit 128, cusp cap 3; pose-key bins 10^{-6} m and 10^{-8} rad.
Shared screening	Reverse weight $w_R = 1.35$; one-cusp feasible-candidate cap $K_f = 30$; post-solution budget $\tau_{\text{sel}} = 0.5$ s; J selection tolerance 10^{-9} s; result tie tolerance $\tau_{\text{cmp}} = 10^{-7}$ s; numerical checking allowance $\varepsilon_{\text{num}} = 10^{-5}$ m.
Small-angle handling	$\phi_\epsilon = 10^{-6}$ rad; exact-zero threshold $64\epsilon_{\text{machine}}$.

construction selection in Branches II and III. First and final solution records expose the computation spent after finding a feasible trajectory.

C. Evaluation Measures

For task i, let $\mathcal{S}_i$ contain its successful methods in the comparison being summarized. With J defined in (7), the per-task reference values and relative measures are

$$J_i^{\text{best}} = \min_{m \in \mathcal{S}_i} J_{i,m}, \qquad L_{i,m}^{J} = 100 \frac{J_{i,m} - J_i^{\text{best}}}{J_i^{\text{best}}}, \tag{21}$$

$$t_i^{\text{best}} = \min_{m \in \mathcal{S}_i} t_{i,m}, \qquad L_{i,m}^{t} = 100 \frac{t_{i,m} - t_i^{\text{best}}}{t_i^{\text{best}}}. \tag{22}$$

L^J measures the gap to the best tested valid result; L^t measures relative computation-time overhead. Quality and relative-time averages use the same common-success task set, whose size accompanies each table. Success counts and absolute planning-time means use all attempted tasks. Best-result counts include ties within τ_{cmp}. Pairwise analyses state their own common-success counts. Unpenalized T_f, extra stops N_s, and stationary time expose the components of J.

Each successful final trajectory undergoes independent checking at 0.02 m spacing, including exact terminal tolerances, the final reverse straight, body clearance, curvature, and junction consistency. The time profile separately checks acceleration, braking, lateral acceleration, and steering rate. The release includes failed and timed-out calls as well as successful trajectories.

D. Overall Results and Mechanisms

Table IV reports 89 common-success tasks. BRS solves all 100 tasks within the planning deadline and attains the lowest tested J on 51 tasks, including ties. Its mean model-predicted execution time is 67.84 s, with 0.47 extra stops and a mean per-task cost gap of 9.67%. It also obtains the smallest unpenalized execution time in 50 tasks.

The representation-controlled comparison uses the same continuous-curvature connector in BRS and CC-A*. On the common tasks, BRS has lower J in 33 cases and CC-A* in 49; the others tie. Their mean costs are 72.56 and 83.64 s. The two methods place intermediate states and cusp candidates differently. BRS uses endpoint-specific fan geometry and staged continuation, whereas CC-A* explores pose bins. The common curve family permits these candidate-organization strategies to be compared with the same steering continuity.

HA has mean moving and stationary execution times of 61.26 and 25.96 s; BRS gives 66.66 and 1.18 s. These components show how steering stops influence execution independently of geometric path length. The common J further accounts for the specified extra-stop preference.

BRS obtains its first complete feasible trajectory in 0.338 s on average. Its mean final planning time is 0.411 s, median 0.013 s, 95th percentile 2.223 s, and maximum 8.088 s. The direct family returns promptly, while difficult queries spend more work on collision screening and connector generation before prefix continuation. Figure 6 shows the per-task quality and time distributions over all 100 tasks; unsuccessful queries remain outside the successful fractions.

Figure 7 examines Case #3. CC-A* selects a continuous-curvature trajectory with predicted execution time 28.24 s, compared with 37.74 s for BRS. Both avoid additional stops. The circular-primitive outputs require 3, 4, and 3 internal stops for HA, BIAGT, and GHA-search. The decomposition separates moving time from stationary steering and gear adjustment.

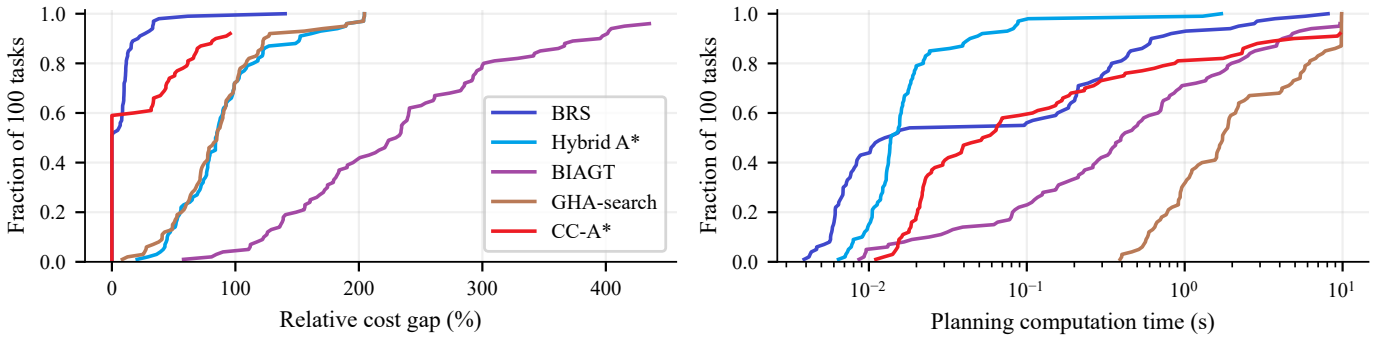
E. Extended First-Solution Diagnostics

The 12 task–method pairs without a valid result under the main deadline are additionally run with a 500 s diagnostic

TABLE IV
TRAJECTORY QUALITY AND PLANNING TIME UNDER THE COMMON DEADLINE

Method	Success	$\overline{T_f}$ (s)	$\overline{N_s}$	$\overline{N_c}$	$\overline{J}$ (s)	$\overline{L^J}$ (%)	Best count	$\overline{t}$ (s)	Median t (s)	$\overline{L^t}$ (%)
BRS	100/100	67.84	0.47	0.47	72.56	9.67	51	0.411	0.013	2044.1
HA	100/100	87.21	3.79	0.58	125.08	95.71	0	0.050	0.014	82.1
BIAGT	96/100	132.54	8.29	2.18	215.46	231.77	0	1.471	0.399	12579.4
GHA-search	100/100	86.09	3.58	0.51	121.93	90.85	0	3.309	1.738	22607.2
CC-A*	92/100	79.15	0.45	0.45	83.64	20.24	59	1.315	0.055	3182.6

Quality and relative-time means use the 89 common-success tasks. Absolute time means include all 100 attempts. Best counts include ties.



(a) Relative trajectory cost (b) Successful planning time

Fig. 6. Per-task cost gaps and planning runtimes under the common vehicle model.

TABLE V
COMPONENTS WITH THE SAME QUERY DEADLINE AND VEHICLE MODEL

Configuration	Success	ΔJ (s)	$\bar{t}$ (s)	Feasible count
I + II + prefixes	99/100	-0.83	0.507	8.4
I + III + prefixes	98/100	-0.42	0.378	10.4
Fixed II–III order	100/100	-1.37	0.451	9.9
Bounded refinement	100/100	-0.19	0.395	9.5
BRS	100/100	0	0.411	9.4

ΔJ uses paired successes. Feasible count denotes complete candidates evaluated per query.

limit. Each diagnostic returns its first complete feasible trajectory. Figure 8 compares first-solution times on the affected tasks. 12 of these 12 calls obtain a valid trajectory within the diagnostic limit. Their first-solution times range from 10.72 to 44.90 s. The main success rates and quality comparison retain the original 10 s outcomes.

F. Component Analysis

The single-sided configurations retain Branch I and the same prefix mechanism. Goal-side-only planning solves 99 tasks; start-side-only planning solves 98. Their complementary failures show why both cusp constructions are retained. Fixed II–III ordering solves 100 tasks, with mean planning time 0.451 s and paired mean cost change -1.37 s relative to BRS. Its mean connector-generation count is 242.9, compared with 233.8 for the heading-prioritized order.

Figure 9 compares first-feasible selection with (K_f, τ_{sel}) equal to $(3, 0.1)$, $(10, 0.25)$, and $(30, 0.5)$, with times in seconds. All four configurations solve 100 tasks within the query deadline. The selected configuration reduces mean J from 77.43 to 75.11 s while increasing mean planning time from 0.332 to 0.395 s. It improves 40 tasks and worsens 0. Branch I remains immediate-return throughout. Optional bounded refinement retains the coarse incumbent and solves 100 tasks.

G. Representative Constructions and Continuation

Figures 4 and 5 separate the actual sampled states, tested curves, and selected motions. The two fan constructions place cusps using the available space at opposite endpoints.

The local constructions solve 95 tasks from their original starts. Prefix continuation solves the remaining 5 tasks: Cases #19, #22, #27, #30, #100. Figure 10 shows Cases #27 and #30. The former becomes reverse-connectable after one prefix. The latter uses two prefix motions to reach a pose from which a goal-side entry succeeds. Each child retains the original goal, footprint, terminal approach, and shared deadline; its full trajectory is checked from the original start.

H. Code and Data Availability

The native C++ BRS implementation, 100 benchmark tasks, and all four comparison planners are available at https://github.com/libai1943/MiningPocket. The repository includes fixed parameters, build and experiment scripts, all recorded outputs, the common evaluator, and independent validation. Reviewers can inspect and rerun the complete comparison under the stated task definition.

IX. CONCLUSION

BRS organizes three complementary motion constructions and bounded recursive prefixes for narrow unstructured mining loading pockets. Continuous-curvature interfaces support rate-limited execution. Immediate reverse-only return

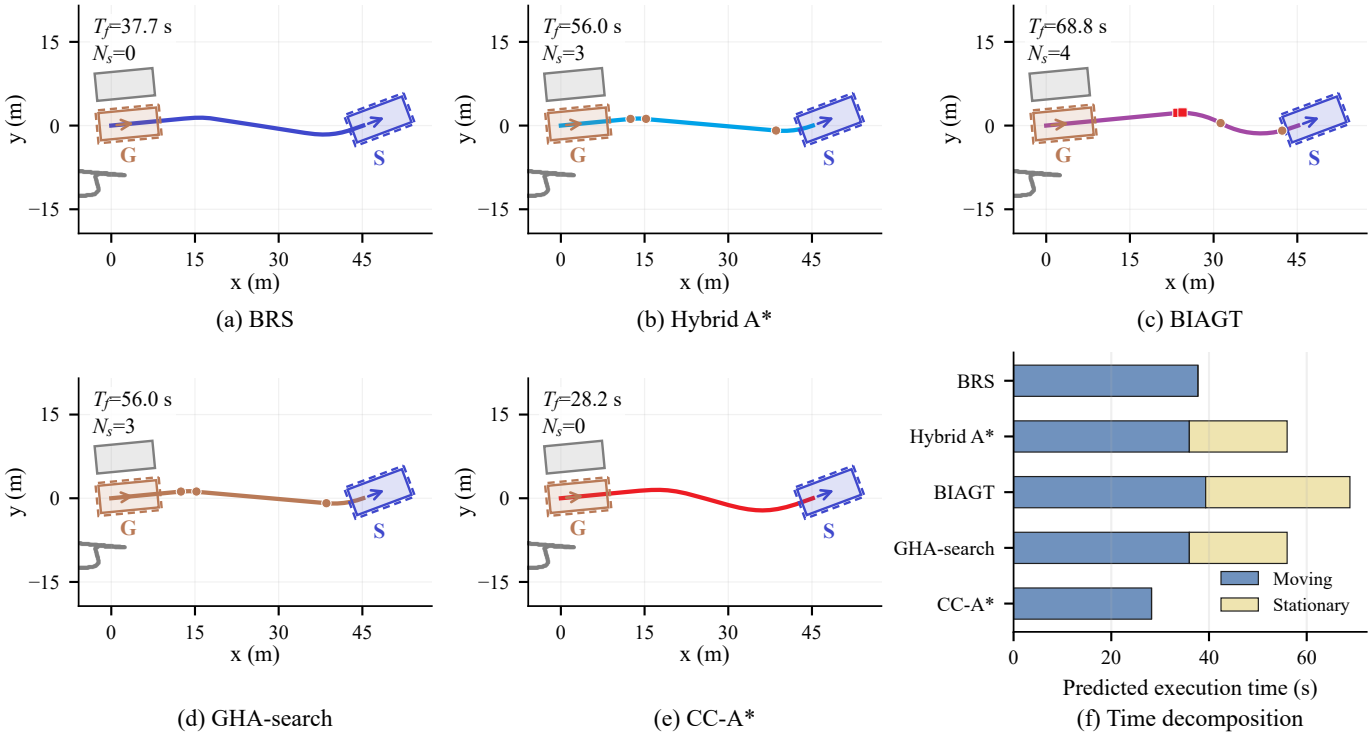


Fig. 7. Five planners on one loading-pocket task, with gear-change and stationary-steering events.

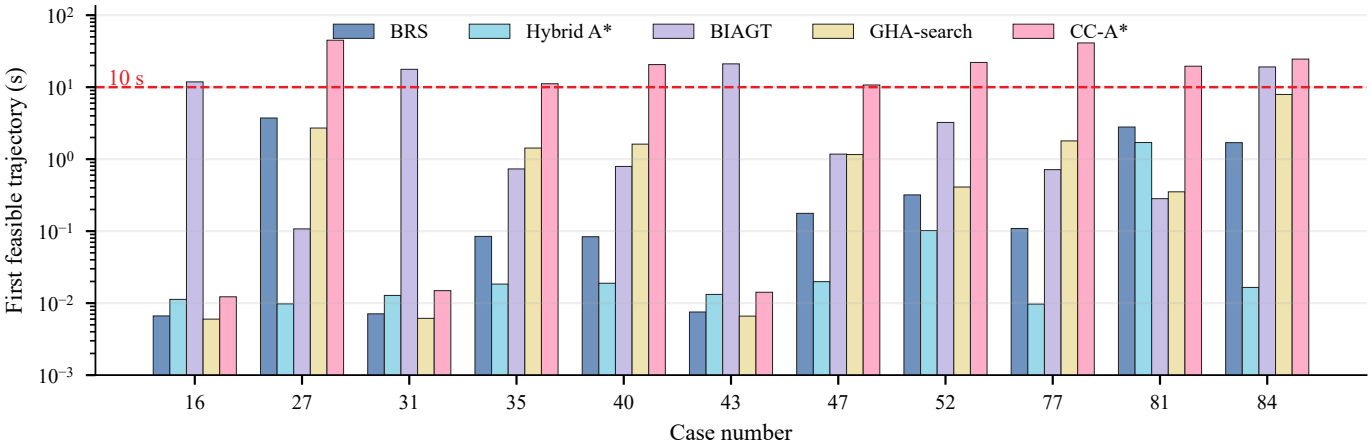


Fig. 8. First-solution times on tasks requiring extended diagnostics for at least one planner.

and bounded one-cusp selection balance planning work and complete-trajectory quality. The native C++ implementation solves all 100 benchmark tasks within 10 s, with mean planning time 0.411 s and maximum 8.09 s. It attains the lowest tested cost on 51 tasks including ties and a mean per-task cost gap of 9.67% over the common-success set. The shared-representation search comparison and component analysis connect these results to candidate placement, prioritized exploration, and prefix continuation.

The theoretical results concern finite candidate selection, a specified lazy search graph, and bounded prefix coverage under explicit assumptions. Future work can study stronger continuous-space coverage properties, additional field maps, and measured truck execution.

Appendix A
Selection Properties and Scope of Coverage

Proposition 1 (Ordered-catalogue selection)**.** *For a finite complete-candidate catalogue ordered by C_{geo}, with terminating feasibility checks, the first accepted candidate minimizes C_{geo} over its feasible members. This also holds for lazy fan expansion when each unresolved queue key lower-bounds every candidate it can generate.*

Proof. Earlier checked candidates are infeasible; later complete candidates have no smaller cost. Every unresolved fan has a key no smaller than the popped complete candidate and lower-bounds all its completions. It therefore cannot contain a cheaper feasible candidate. □

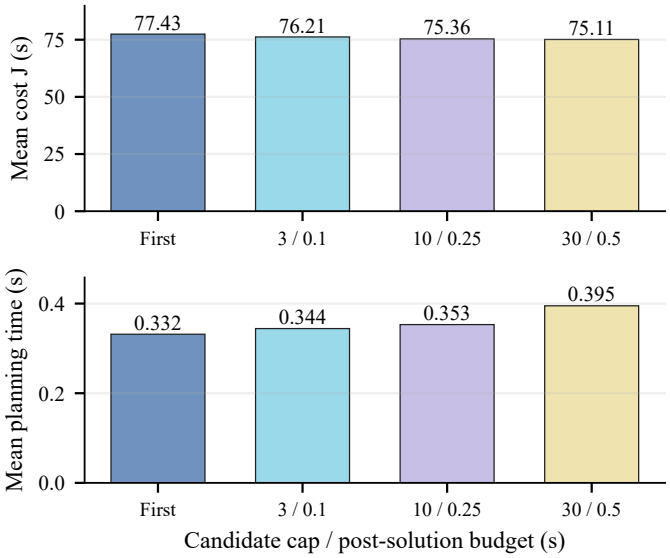


Fig. 9. Bounded candidate selection trades additional planning time for lower complete-trajectory cost.

Proposition 2 (Execution-cost retention)**.** *Let $\mathcal{Q}$ contain the complete validated trajectories actually compared within the successful construction, including any explicitly invoked refinement. Under deterministic common time parameterization and comparison tolerance τ_J, the returned cost is at most $\min_{q\in\mathcal{Q}} J(q)+\tau_J$. Further incumbent-preserving comparisons cannot increase it.*

Proof. Replacement requires an improvement exceeding τ_J. Each skipped candidate costs at least the incumbent minus τ_J, and subsequent replacements only reduce that incumbent. Prefix–suffix candidates are scored as complete trajectories before replacement, so the same argument applies to child invocations. □

Proposition 3 (Lazy finite-DAG minimum)**.** *Fix a finite DAG and finite deterministic candidate lists. Assume (i) unknown edge weights lower-bound candidate lengths; (ii) edge lists are length-sorted; (iii) feasibility checks are consistent; and (iv) search fully validates a minimizing route before exhausting its evaluation budget. The accepted route then has minimum length among feasible routes represented by these lists.*

Proof. Every edge weight lower-bounds its shortest feasible candidate: initialization ensures (i), generation preserves the bound by (ii), and rejection removes only infeasible candidates by (iii). Bellman's recurrence consequently lower-bounds every feasible complete route. Once its minimizing route is fully validated, each selected edge weight equals its accepted length, attaining that bound. □

Finite lists allow only finitely many generation/rejection updates. Root finding and deduplication define these fixed lists. A budget-limited refinement retains its coarse incumbent; Proposition 2 governs subsequent execution-cost selection.

Proposition 4 (Termination and conditional prefix coverage)**.** *With finite primitive lengths, four gear/steering modes, finite local catalogues, finite depth and state caps, and terminating numerical routines, BRS terminates. Under exhaustive expansion through depth d, nonbinding time/state caps, and no merging of continuation-inequivalent states, every checker-accepted prefix sequence of depth at most d satisfying the cusp cap is tested unless a solution returns earlier.*

Proof. With $b = 4|\mathcal{L}_p|$ choices per state, the unmerged tree has at most $1+b+\cdots+b^d$ states. Every local invocation and prefix check is finite; the implemented caps further limit work. For exhaustive coverage, induct on depth: the root is processed, and every processed parent generates all checker-accepted children respecting the cusp cap. Breadth-first scheduling processes each retained child. The merging condition preserves every necessary continuation; earlier success satisfies the existence objective. □

Implemented pose-key merging and budgets restrict this coverage. Terminal success additionally requires a feasible local connector.

A. Small-Angle Regularity

Within the regularization range, $\sigma = O(|\phi|^{1/2})$, $k_p = O(|\phi|^{3/4})$, and $L_r = O(|\phi|^{1/4})$. Length, curvature, and sharpness therefore vanish as $\phi \to 0$. Since $|\delta'| \le L_w\sigma$, the steering-rate ceiling becomes inactive. Exact zero-turn roots and unmodified numerical roots preserve this limiting geometry.

B. Continuous Coverage and Hybrid A*

Continuous-space resolution completeness requires coordinated refinement of primitive lengths, steering profiles, connector sampling, and budgets, with pruning preserving positive-clearance solutions. Existing lattice and sampling guarantees likewise depend on coverage assumptions [11], [15].

BRS prefixes use curvature ramps, bounded child-local selection, and keys including gear, depth, and cusps. Hybrid A* typically retains one representative continuous pose per cell and expands designated steering primitives. Establishing coverage equivalence without RS closures would require corresponding action sets, state merging, and terminal tests. The verified connector supplies terminal accuracy here; any continuous-space guarantee must preserve the collision, endpoint, and cusp constraints.

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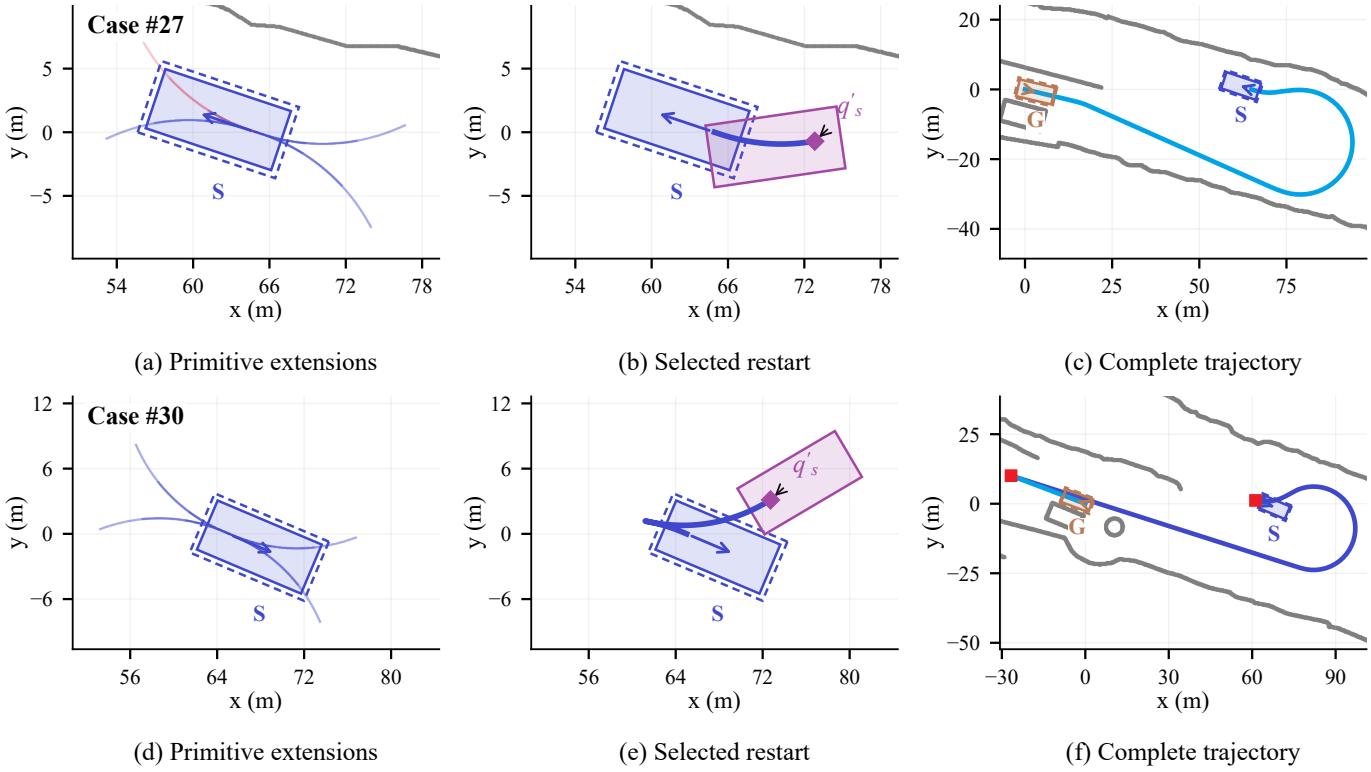


Fig. 10. Reachable prefixes change the starting state and enable a complete validated continuation.

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